

VictorAmaya

Enterprise AI Engineer · Industrial Intelligence · GPU · Full-Stack

Real-world AI for systems that already run the world.



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Enterprise AI engineer with **12+ years** shipping production software for industrial manufacturing, AI/simulation R&D, financial exchanges, and enterprise XR. Currently building **Ciro AI**, a production AI platform deployed across three countries for chemical and pharmaceutical manufacturers — real-time computer vision, multi-agent LLM systems, and GPU-backed inference on the factory floor. Earlier led ML and simulation systems at Vertex Studio (\$1M+ ARR), shipped enterprise XR products at VirseHQ (\$15M ARR), and built matching-engine infrastructure at **Kraken**.

CORE EXPERTISE

AI / ML Systems	10+ years Production ML, GPU inference, computer vision, MLOps (PyTorch, TensorRT)
LLMs & Agents	5+ years Multi-agent systems, RAG, tool-use, NL-to-SQL (OpenAI, Anthropic, LangChain)
Computer Vision	10+ years Industrial QC, defect detection, real-time inference on edge (YOLO, DETR, OpenCV, TensorRT)
GPU & CUDA	10+ years Custom kernels for inference, training, and proving; H100 / Jetson deployment
TypeScript / React	10+ years Web platforms, dashboards, SDKs (Next.js, Vite, Three.js, React Native)
Node.js / Python	12+ years APIs, real-time backends, ETL pipelines, FastAPI, GraphQL
Rust / C / C++	12+ years High-performance services, matching engines, low-latency systems
Cloud & DevOps	10+ years AWS, Docker, Kubernetes, Terraform, MLflow, observability

EXPERIENCE

Founding Engineer & CTO — **Ciro AI**

Aug 2024 — Present · Remote · US / LATAM

- Built the full **Ciro** platform end-to-end: **TypeScript/React** dashboards, **Node.js** + **FastAPI** backends, **PostgreSQL** + **DuckDB** analytical store, **Python** ML services. Deployed in production across three countries.
- Architected GPU-backed agent system: **LLMs**, transformer-based vision models (YOLO, DETR), real-time anomaly detection, NL-to-SQL over operational data.
- Designed and shipped the **Orbita** field-sales mobile app: **React Native** + **Expo**, offline-first sync, conflict resolution, deployed to manufacturing distributors.
- DTE/DGII tax-integration layer for LATAM ERP customers: **Node.js** services, async queues, audit trails. XFounders Accelerator (Starknet & StarkWare); presented at Hi Tech! Stuttgart.

Founding Engineer — BitSage Network — Obelyzk Protocol

Sep 2025 — Present · Remote

- Authored **Obelyzk**: open-source Rust ZK proving engine for ML. Verified Qwen2.5-14B (14B params, 281 transformer layers) and GLM-4-9B on Starknet Sepolia.
- Built the first GKR sumcheck prover on StarkWare's **STWO** (Circle STARKs, M31 field) — first on-chain recursive STARK verification of ML inference.
- **20000+ lines of custom CUDA** (MLE restriction, sumcheck, Poseidon Merkle). **950+ tests**. Published as **obelyzk** on crates.io / PyPI, **@obelyzk/sdk** on npm.
- Built the **Next.js** marketplace + docs site, **TypeScript** SDK, and Discord bot integrations. Speaker, **zkSummit 14**, Rome (May 2026).

Lead Protocol Engineer — SugarFunge Protocol

Jan 2022 — Aug 2024 · Las Vegas, NV

- Designed on-chain behavioral-economics engine on **Substrate / Rust** for high-frequency trading and game economies. Trustless matching, settlement, asset bundling.
- Built marketplace web app (**Next.js / TypeScript / GraphQL**), Discord and Telegram bot integrations, on-chain analytics dashboard.

Lead ML & Simulation Engineer — Vertex Studio

Mar 2020 — Mar 2023 · California

- Led ML and simulation systems for enterprise synthetic-data generation; product grew from R&D to **\$1M+ ARR**. Stack: **Python / PyTorch, Unreal Engine**, custom rendering pipelines.
- Shipped the web platform in **React + TypeScript**, with a **Node.js** job-orchestration layer and AWS Batch for distributed rendering. Integrated WebGL preview using Three.js.

- Computer-vision pipelines (PyTorch, OpenCV, TensorRT) for production deployment on NVIDIA edge devices.

Product Engineer — Kraken (via skalex GmbH) Mar 2017 — May 2020 · Munich, Germany

- Matching engines, liquidity protocols, and HFT infrastructure for tier-one cryptocurrency exchange operators. **Node.js + TypeScript** services, **C++** low-latency components, Redis for hot-path state.
- High-availability architecture: active-active deployment, custom WebSocket fan-out, fault-tolerant order books with strict latency budgets.

Product Engineer & Regional Director — VirseHQ Jan 2015 — Mar 2017 · Los Angeles

- Full-stack on AI-powered metaverse / XR enterprise products (**\$15M ARR**). **React, Node.js, Unity / Unreal**, WebGL clients, real-time multiplayer networking.
- Built 3D content pipelines, real-time rendering tooling, and Meta-platform integrations for enterprise XR deployments.

Chief Innovation Officer — Novatech / Kadevjo Jan 2014 — Dec 2015 · San Salvador, El Salvador

- Co-founder, game development studio. Shipped multiple mobile titles (Unity / C#) and early crypto integrations. Grew to **\$1M ARR**, exited.

EDUCATION

MIT Sloan School of Management — MicroMasters — Finance 2024 — 2025
Stanford University — Game Theory & Economics 2021 — Present
Harvard University (Coursera) — CS50 Python · AI Product Management

SKILLS

ML / AI	PyTorch, TensorFlow, JAX, CUDA, ONNX, TensorRT, Hugging Face, distributed training, fine-tuning (LoRA, QLoRA)
LLMs / Agents	OpenAI APIs, Anthropic APIs, LangChain, LlamaIndex, vLLM, RAG pipelines, tool-use agents, function calling, NL-to-SQL
Vision	YOLO, DETR, Segment Anything, OpenCV, NVIDIA DeepStream, GStreamer, FFmpeg, synthetic-data generation
Languages	Python, TypeScript, JavaScript, Rust, Node.js, C/C++, Go, Swift, C#, Lua, Cairo, Solidity
Frontend	React, Next.js, Vite, Three.js, WebGL, React Native, Expo, Tailwind, Framer Motion
Backend	Node.js, FastAPI, Express, GraphQL, REST, PostgreSQL, Redis, DuckDB, Apache Parquet, Kafka, gRPC
3D / Sim	Unreal Engine, Unity, Blender, OpenGL, GLSL, NVIDIA PhysX
Infrastructure	AWS, GCP, Docker, Kubernetes, Terraform, MLflow, Weights & Biases, GitHub Actions, OpenTelemetry, Grafana

SELECTED WORK & TALKS

Obelyzk.rs — open-source ZK proving engine (crates.io/crates/obelyzk). • **On-chain verification** of Qwen2.5-14B & GLM-4-9B on Starknet Sepolia (Apr 2026). • **zkSummit 14, Rome** — “A Practitioner’s Guide to GKR Sumcheck Over Circle STARKs” (May 2026). • **ETHGlobal Cannes** hacker — Verifiable AI Agent for On-Chain Fund Management (Apr 2026). • **Hi Tech! Stuttgart Match Arena** Batch #9 (Nov 2025).